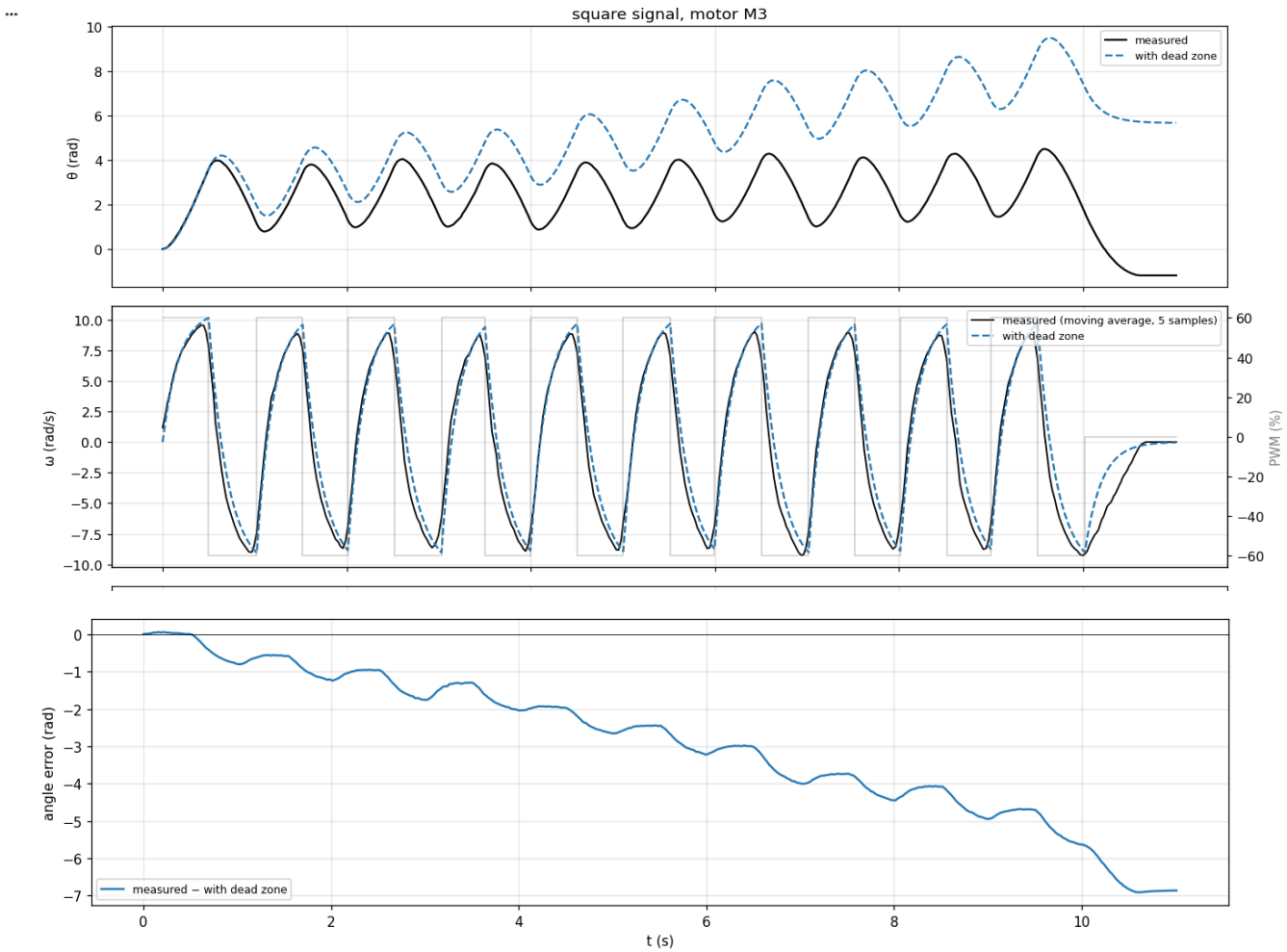
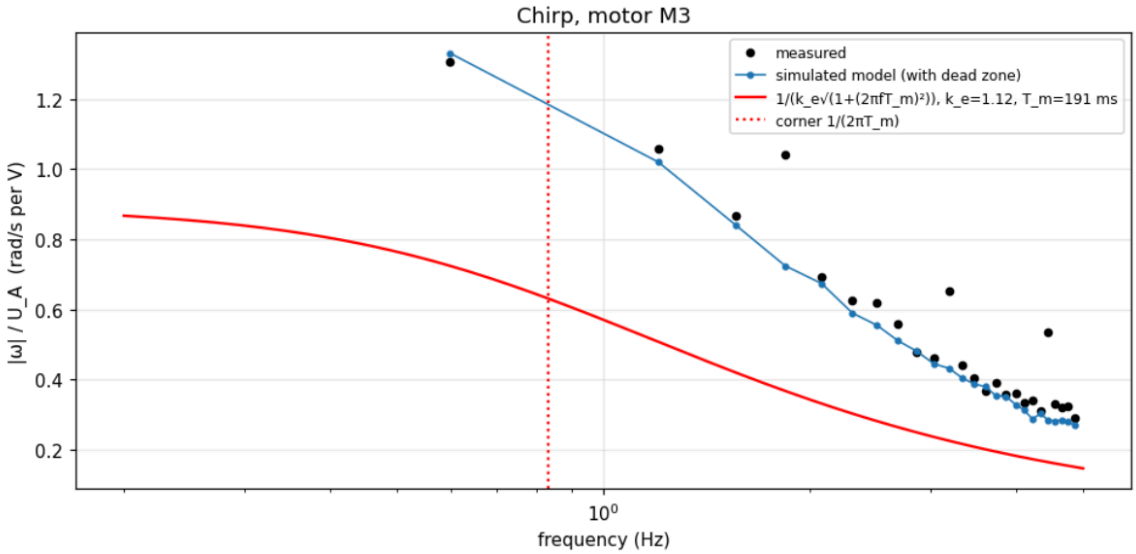


1) The blue dashed line in the first graph is the predicted mathematical model (for square signal) of the encoder angle across time derived using the parameters found by the step model.And the black graph is the motor's measured angle under that input in that time interval. The predicted model predicts the reversals correctly but it predicts the encoder angle values very poorly.

2)IN the second graph we take the average of the measurements in each 5 overlapping intervals. In this graph the actual and predicted models align almost perfectly but with very small deviations which can add up to encoder angle error when we calculate the area under velocity (integral).

3) The error between predicted and actual model. The error becomes increasingly **negative** as the predicted angle is higher than the measured angle.

Identical results for M3. Other graphs can be explained in the same way but with different errors and deviations.



$$\omega_A = \frac{U_A}{k_e \sqrt{1 + (2\pi f T_m)^2}}$$

The y-axis is motor speed amplitude divided by input voltage amplitude. It tells you how much speed amplitude the motor produces per volt of input. Higher y-value = the motor reaches higher forward and backward speed values.

Frequency counts the complete command cycles per second : In our case how many times the motor reaches from the lowest to the highest PWM values in the given window (example: from -60 to 60). The higher the frequency gets, the time to reach high speeds shrink, so the input changes the pwm value before reaching the high speed values.

So from the black dots (They are coming from the actual motor) we can see that the smaller the frequency we have(moving left) the larger our speed swing values. When we increase the frequency the motor reaches smaller speed swings. The blue line is the mathematical prediction for chirp signal, and it aligns very good with the actual chirp values. The blue line adds the offset found by the step model but the red line doesn't. That is why they show big difference.

Which prediction better:

The model predicts speed best for the triangle input, with fit values of 85.26% for M3 and 84.26% for M4. The square input gives lower fits of 77.66% and 78.45%, respectively. The

chirp input gives the lowest fits: 65.77% for M3 and 58.14% for M4. Although the square-input speed curves follow a similar pattern, systematic speed errors accumulate into a large angle drift.

With and without dead the model shows identical behaviour but shifted.